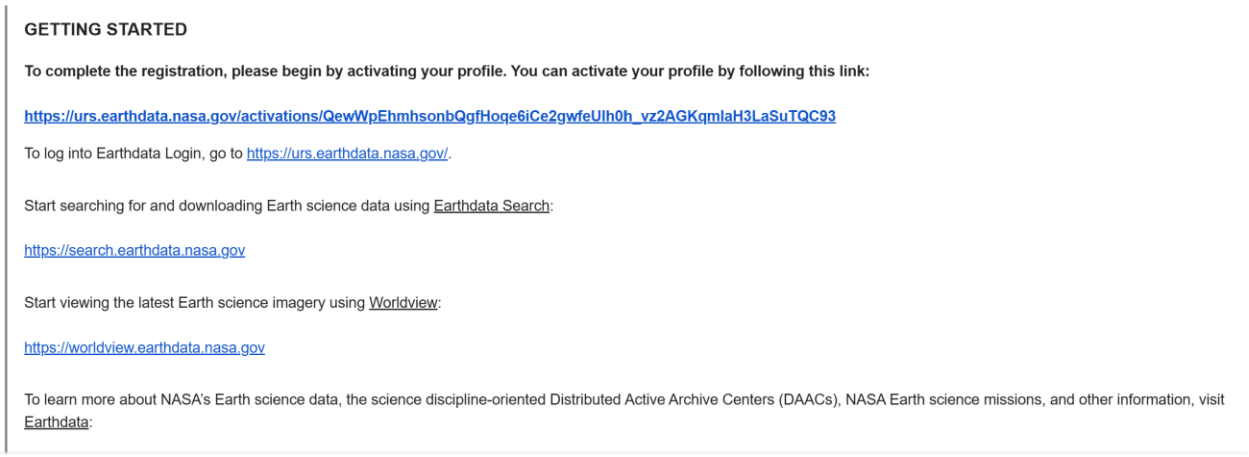
A. NASA EarthData Setup (5- 10 minutes)

1. Sign Up for Earthdata Account:
   - Go to https://urs.earthdata.nasa.gov/users/new
   - Register for an account and verify your email.



GETTING STARTED

To complete the registration, please begin by activating your profile. You can activate your profile by following this link:

https://urs.earthdata.nasa.gov/activations/QewWpEhmhsonbQgfHoqe6iCe2gwfeUlh0h_vz2AGKqmlaH3LaSuTQC93

To log into Earthdata Login, go to https://urs.earthdata.nasa.gov/.

Start searching for and downloading Earth science data using Earthdata Search:

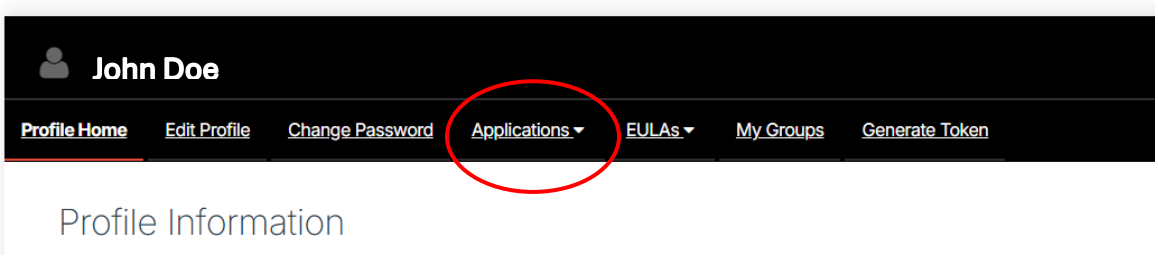
https://search.earthdata.nasa.gov

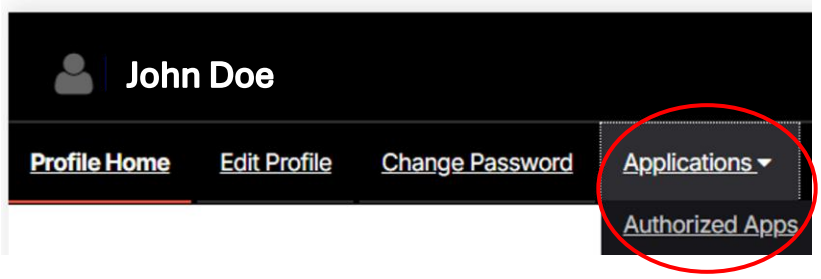
Start viewing the latest Earth science imagery using Worldview:

https://worldview.earthdata.nasa.gov

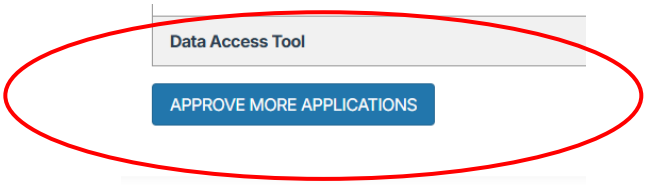
To learn more about NASA's Earth science data, the science discipline-oriented Distributed Active Archive Centers (DAACs), NASA Earth science missions, and other information, visit Earthdata:

2. After account development, users will need to authorize applications. After logging in from your profile page, Go to applications, select Authorized Apps

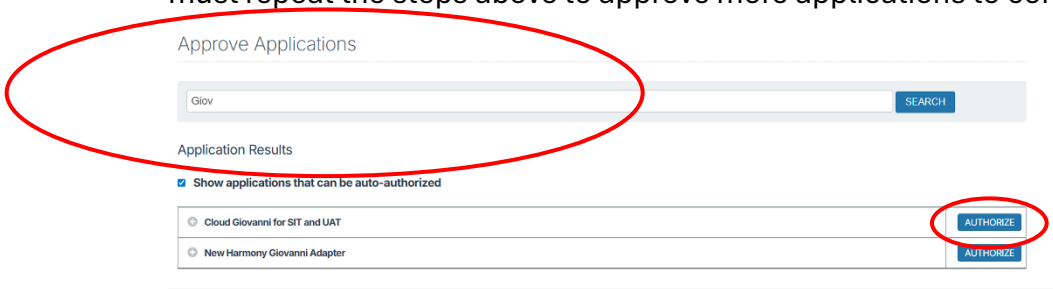
3. 



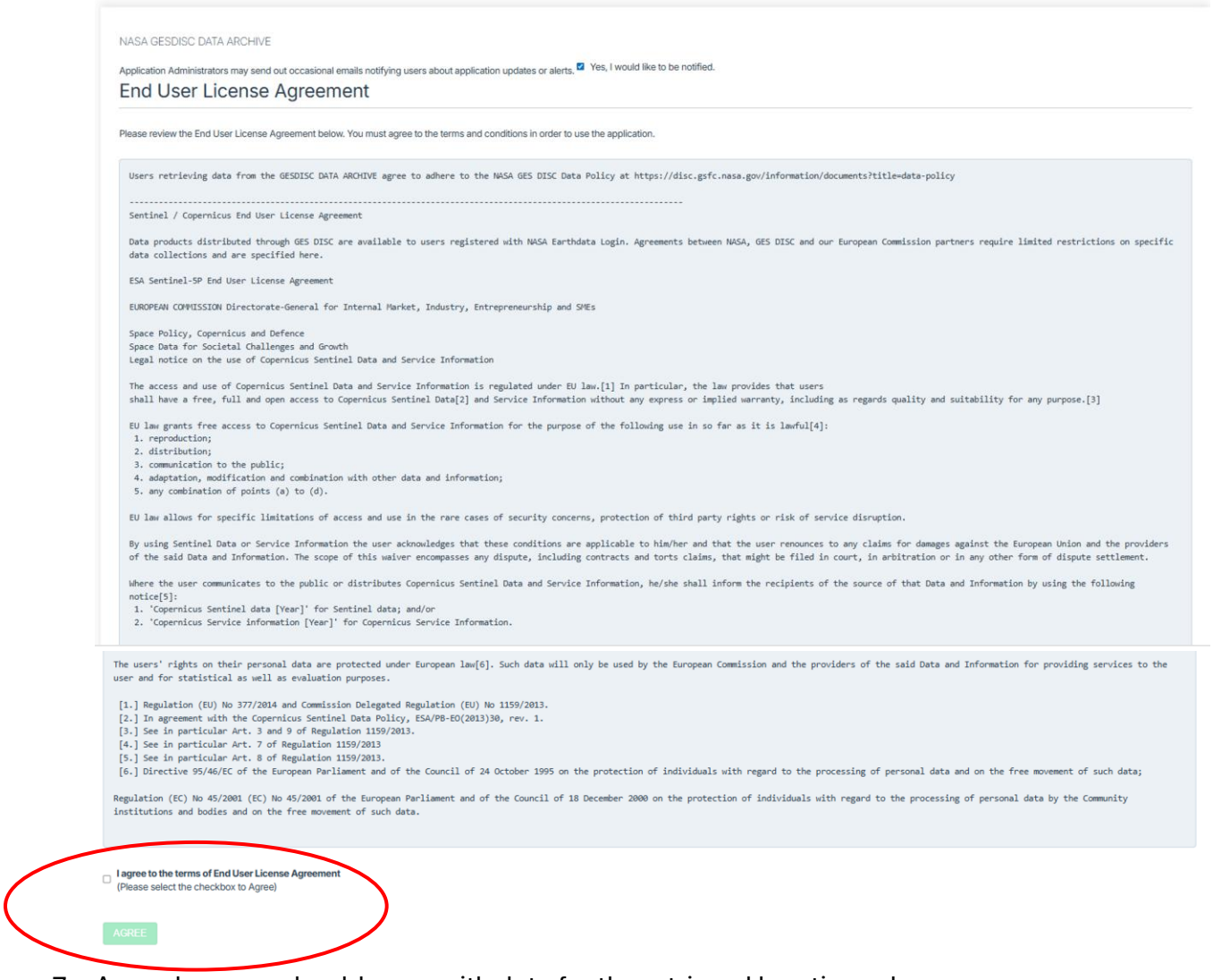
4. On the page, Click Approve More Applications



5. In the search box, type in Giov: add each Giovanni products available: Main 2 are Giovanni, Cloud Giovanni (Test). Click Authorize to the right of the application. You must repeat the steps above to approve more applications to continue.



6. New Users go to https://api.giovanni.earthdata.nasa.gov/proxy-timeseries?data=NLDAS_FORA0125_H_2_0_Rainf&location=[46.6875,-93.0625]&time=2010-01-01T01:00:00/2025-10-24T23:00:00&version=2.0 to ensure the data connection has been made. If new to Giovanni, you will need to accept the end user agreement to the NASA GESDISC DATA Archive. A new browser page will display as seen below. Important – If the data doesn't appear after accepting the user agreement, you may need to wait for the NASA server to update your user preferences from the Giovanni approved applications.



NASA GESDISC DATA ARCHIVE

Application Administrators may send out occasional emails notifying users about application updates or alerts. ☑ Yes, I would like to be notified.

End User License Agreement

Please review the End User License Agreement below. You must agree to the terms and conditions in order to use the application.

Users retrieving data from the GESDISC DATA ARCHIVE agree to adhere to the NASA GES DISC Data Policy at https://disc.gsfc.nasa.gov/information/documents?title=data-policy

---

Sentinel / Copernicus End User License Agreement

Data products distributed through GES DISC are available to users registered with NASA Earthdata Login. Agreements between NASA, GES DISC and our European Commission partners require limited restrictions on specific data collections and are specified here.

ESA Sentinel-5P End User License Agreement

EUROPEAN COMMISSION Directorate-General for Internal Market, Industry, Entrepreneurship and SMEs

Space Policy, Copernicus and Defence
Space Data for Societal Challenges and Growth
Legal notice on the use of Copernicus Sentinel Data and Service Information

The access and use of Copernicus Sentinel Data and Service Information is regulated under EU law.[1] In particular, the law provides that users shall have a free, full and open access to Copernicus Sentinel Data[2] and Service Information without any express or implied warranty, including as regards quality and suitability for any purpose.[3]

EU law grants free access to Copernicus Sentinel Data and Service Information for the purpose of the following use in so far as it is lawful[4]:

1. reproduction;
2. distribution;
3. communication to the public;
4. adaptation, modification and combination with other data and information;
5. any combination of points (a) to (d).

EU law allows for specific limitations of access and use in the rare cases of security concerns, protection of third party rights or risk of service disruption.

By using Sentinel Data or Service Information the user acknowledges that these conditions are applicable to him/her and that the user renounces to any claims for damages against the European Union and the providers of the said Data and Information. The scope of this waiver encompasses any dispute, including contracts and torts claims, that might be filed in court, in arbitration or in any other form of dispute settlement.

Where the user communicates to the public or distributes Copernicus Sentinel Data and Service Information, he/she shall inform the recipients of the source of that Data and Information by using the following notice[5]:

1. 'Copernicus Sentinel data [Year]' for Sentinel data; and/or
2. 'Copernicus Service information [Year]' for Copernicus Service Information.

The users' rights on their personal data are protected under European law[6]. Such data will only be used by the European Commission and the providers of the said Data and Information for providing services to the user and for statistical as well as evaluation purposes.

[1.] Regulation (EU) No 377/2014 and Commission Delegated Regulation (EU) No 1159/2013.
[2.] In agreement with the Copernicus Sentinel Data Policy, ESA/PB-EO(2013)30, rev. 1.
[3.] See in particular Art. 3 and 9 of Regulation 1159/2013.
[4.] See in particular Art. 7 of Regulation 1159/2013
[5.] See in particular Art. 8 of Regulation 1159/2013.
[6.] Directive 95/46/EC of the European Parliament and of the Council of 24 October 1995 on the protection of individuals with regard to the processing of personal data and on the free movement of such data;

Regulation (EC) No 45/2001 (EC) No 45/2001 of the European Parliament and of the Council of 18 December 2000 on the protection of individuals with regard to the processing of personal data by the Community institutions and bodies and on the free movement of such data.

☐ **I agree to the terms of End User License Agreement**
(Please select the checkbox to Agree)

AGREE

7. A new browser should open with data for the retrieved location url.

B. Develop coding environment for notebook deployment for Data Retrieval

1. To run the notebook, python or ArcGIS (Desktop/Pro) must be installed on the computer.

To install python on the computer please go to:

- Download and install Python 3.x from https://www.python.org/downloads/
- During installation, check the box to "Add Python to PATH".

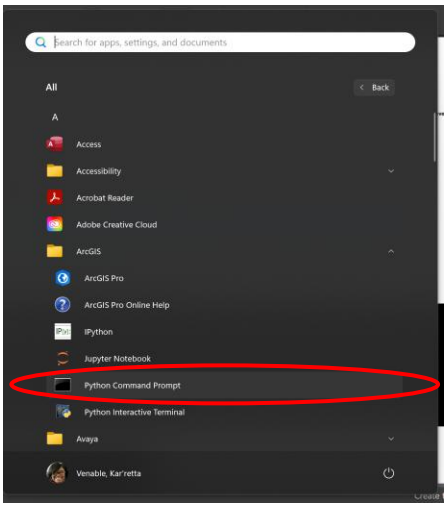
----------Or ----------

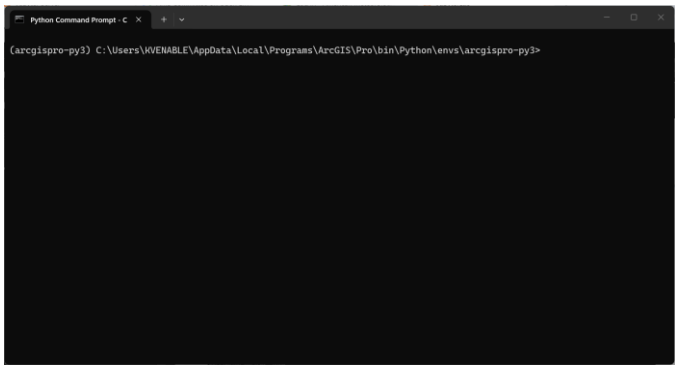
- Install Python and Conda (If Arc GIS is not installed):

- Download and install Anaconda (https://www.anaconda.com/products/distribution)
- Follow the installer prompts.
- Open Anaconda Prompt:

  -Search for "Anaconda Prompt" in your Start menu and open it.
  ----------Or ----------

- If using ArcGIS, click windows button for searching all programs, view all, then scroll to ArcGIS, Click the drop-down menu and select Python Command Prompt





4. Download github repository with jupyter notebook, requirements.txt, and yaml into a folder directory (insert link)

C. Creating the environment and deploying notebooks (Code in bold)

<u>Pip Installation</u>

5. Navigate to Your Project Folder where the github repo was downloaded from the command line prompt:

   cd "C:\Users\JDOE\OneDrive"

6. Go to the opened command line terminal (step 3) and create the environment named EA:
   python -m venv EA
7. Activate the virtual env

cd "C:\Users\JDOE\OneDrive\EA\Scripts\"
C:\Users\JDOE\OneDrive\EA\Scripts\> **activate.bat**
(EA) C:\Users\ JDOE\OneDrive\EA\Scripts\EA\Scripts>**cd ..**
(EA) C:\Users\ JDOE\OneDrive\EA\Scripts\EA\

8. Installed all required packages using the requirements file:
   (EA) C:\Users\ JDOE\OneDrive\EA\>**pip install -r " C:\Users\JDOE\OneDrive\ requirements020426b.txt"**

9. Launch Jupyter Notebook and open the Jupyter notebook file (.ipynb)

python -m jupyterlab

----------Or ----------

<u>If using a conda environment:</u>

5. Navigate to Your Project Folder where the github repo was downloaded from the command line prompt:

cd "C:\Users\JDOE\OneDrive"

6. Go to the opened command line terminal (step 3) and create the environment named EA:
   conda env create -f 020426aeenvironment.yml
7. Activate the virtual env
   conda activate EA
8. Launch Jupyter Notebook:

- Launch Jupyter Notebook and open the Jupyter notebook file (.ipynb)

python -m jupyterlab

----------Or ----------

<u>If using Conda in ArcGIS:</u>

5. Navigate to Your Project Folder where the github repo was downloaded from the command line prompt:

cd "C:\Users\JDOE\OneDrive"

6. Go to the opened command line terminal (step 3) and create the environment named EA:
   conda env create -f 020426aeenvironment.yml
7. Activate the virtual env

activate EA

8. Launch Jupyter Notebook:

- Launch Jupyter Notebook and open the Jupyter notebook file (.ipynb)

python -m jupyterlab

<u>D) Running the notebook</u>

1) The jupyter notebook should be deployed in a new internet browser

2) Run each cell in chronological order

3) To run the cell, click and it should highlight blue then press shift + enter at the same time or click the cell and then navigate to Run on the Ribbon and run selected cell.

4) Most NLDAS data retrievals are available through 12-31-25. Some data products may not have been developed yet for the end of 2025 for some locations and goes back thorough 1/1/1979. Data retrievals are currently available for a maximum of 33 year time intervals.

E) After Environment Setup

After the EarthAccess (EA) environment is created and login credentials are created, there is no need to redevelop the environment or go to the NASA login page. It will only be necessary to activate the environment, deploy the notebook, and have your login credentials for the EA website.